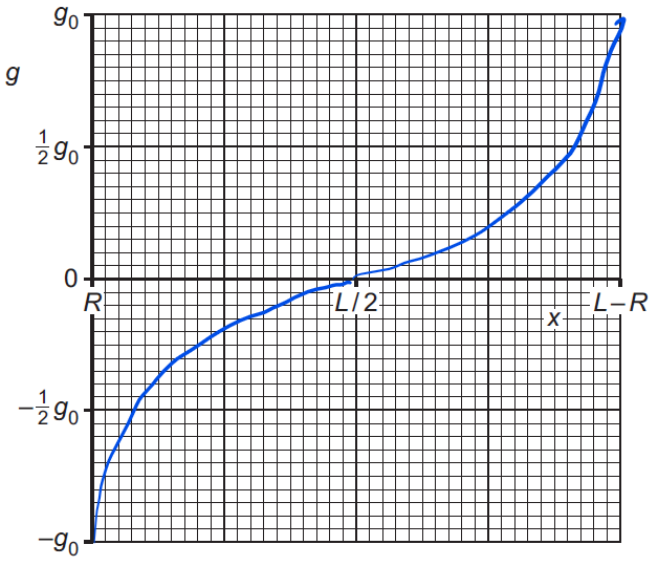
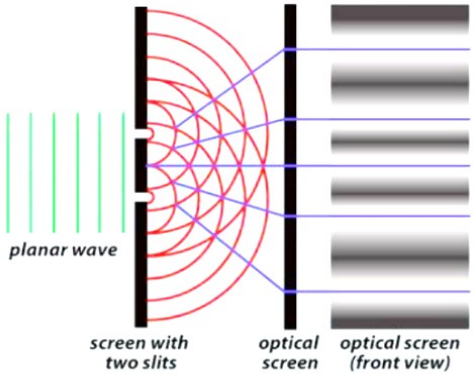
SAJC 2026 H2 Physics Prelim Exams Paper 3 Solutions

1(a)(i)	vertical force of string component = 4.8 sin 30° = 2.4 (N) Taking pivot at hinge, (8.2 × 0.50) = (W × 0.60) + (0.30 × 0.80) + (2.4 × 1.2) W = 1.6 or 1.63 N (2 or 3 sf) (no ecf)	[1] [1] [1]
1(a)(ii)	Resolving force horizontally, horizontal force = 4.8 cos 30° = 4.2 or 4.16 N (2 or 3 sf)	[1]
1(b)	E = ½ Fx 0.32 = ½ (8.2) x x = 0.078 or 0.0780 m (2 or 3 sf) (no ecf)	[1] [1]

2(a)	a = gradient or a = Δv / (Δ)t or a = (v – u) / t a = (0.30 – 0.12) / (0.35 – 0.15) a = 0.90 m s^{-2}	 [1] [1]
2(b)	By conservation of momentum, $P_i = P_f$ →: (0.25 × 0.48) + (0.75 × 0.12) = (0.25 v) + (0.75 × 0.30) or Relative speed of approach = relative speed of separation →: $u_A – u_B = v_B - v_A$ 0.48 – 0.12) = (0.30 – v) v = – 0.060 m s^{-1} or $KE_i = KE_f$ (1/2 × 0.25 × 0.48^2) + (1/2 × 0.75 × 0.12^2) = (1/2 × 0.25 × v^2) + (1/2 × 0.75 × 0.30^2) v = 0.060 m s^{-1} Magnitude of v: 0.060 m s^{-1} Direction: opposite to initial velocity of A / opposite to initial /final velocity of B/ to the left	 [1] [1] [1]

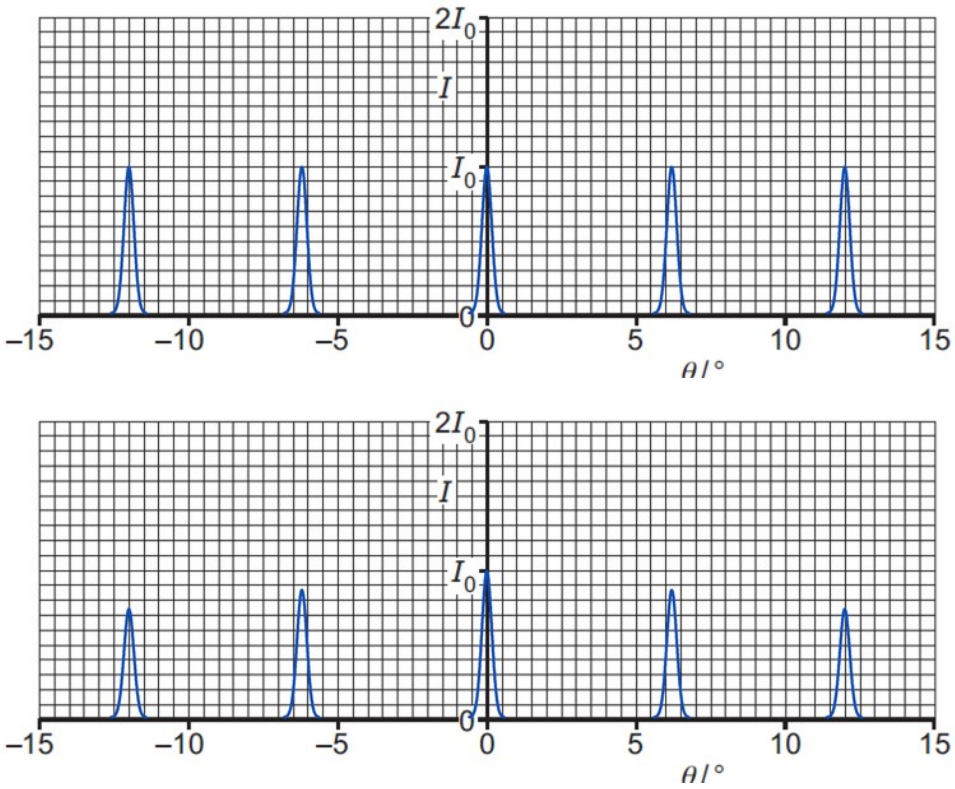
3(a)	Gravitational force provides the centripetal force.	[1]

	$G\frac{Mm}{r^2} = m\frac{v^2}{r}$ $G\frac{Mm}{r} = mv^2$ $\frac{1}{2}mv^2 = \frac{1}{2}\times G\frac{Mm}{r}$ Since $E_P = -G\frac{Mm}{r}$, $E_K = -\frac{E_P}{2}$	[1]
(b)	From graph, $\phi = -3.2\times10^8$ J kg^{-1} at $r = 4.0\times10^8$ m. $E_{tot} = E_K + E_P = -\frac{E_P}{2} + E_P = \frac{E_P}{2} = \frac{m\phi}{2}$ $= \frac{8.93\times10^{22}\times(-3.2\times10^8)}{2}$ $= -1.43\times10^{31}$ J (negative sign essential) or 1. Use data from graph to find mass M of Jupiter using Ep = - GMm/r 2. Use E_T formula E_T = -½ GMm/r 3. Ans = -1.47 x 10^{31} or a bit off	[1] [1, ecf] [1]
(c)	Line starting at (R, $-g_o$) and ending at ($L - R$, $+g_o$) Line passing through (L/2, 0) Curve becomes less steep fr R to (L/2) and then steeper fr (L/2) to ($L - R$)  	[1] [1] [1]

4(a)(ii)	example of transverse waves: light, EM waves, any EM waves example of longitudinal waves: sound }	[1]
4(a)(i)	transverse waves: <u>oscillations/vibrations</u> of <u>wave particles</u> are <u>perpendicular</u> to the direction of <u>energy transfer/transmission/travel</u> longitudinal waves: <u>oscillations/vibrations</u> of wave particles are <u>parallel</u> to the of <u>energy transfer/transmission/travel</u>.	[1] [1]
4(b)(i)	Diffraction refers to the <u>spreading of a wave</u> when it <u>passes through a slit</u> (opening/gap/aperture), or <u>around an obstacle</u> (or around an edge) into the “shadow” region.	[1]
4(b)(ii)	 <u>diffraction/spreading through each slit/slits</u> allows <u>overlap/meet/superpose</u> of secondary wavelets for superposition/interference to occur.	[1] [1]

4(c)(i)	DC =**18 cm**, DABC =**80 cm**	[1]
4(c)(ii)	path difference = 80 -18 = 62 cm = 15½ wavelengths { as λ = 4.0 cm} phase difference =π radians or 180^0 resultant amplitude is a minimum. { Accept: zero }	[1] [1] [1, ecf]

5(a)(i)	$d\sin\theta = n\lambda$ $\theta = \sin^{-1}[(540 \times 10^{-9})/(5.0 \times 10^{-6})]$	[1]

	= 6.2°	
5(a)(ii)	dsin θ = nλ λ = sin^{-1}[2(540 x 10^{-9})/(5.0 x 10^{-6})] = 12.474° = 12° or 12.5°	[1]
5(a)(iii)	peaks/maxima shown at θ = 0, ±6° and ±12° (allow variance of ±1 small sq) peaks/maxima and zero intensity in between (i.e. sharp peaks & flat region in between) central peak/maxima at intensity I_0 and the other peaks all equal to or less than I_0  	[1] [1] [1]
5(b)	intensity (of all peaks / maxima) halved	[1]

6(a)(i)	The battery has internal resistance, so when current flow there is a potential drop across internal resistance.	[1]

	Hence terminal pd/voltmeter reading is less than the emf.	
6(a)(ii)	Given: $R_X = 4\ R_Y$ Deduce $V_Y + V_X = 6.0$ V {Kirchhoff's Voltage Law} Since current is the same for both, $V_Y + 4V_Y = 6.0$ V $V_Y = 1.2$ V	[1] [1]

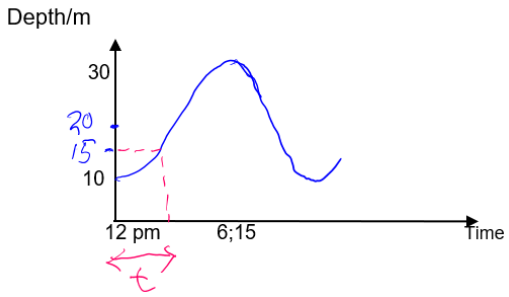
Source: J2000 P3 Q4

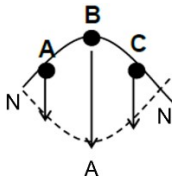
6(b)(i)	Equivalent capacitance of two 50μF capacitors <u>in series</u> = (50 x 50)/ [50 +50] = 25 μF Total capacitance of two 25μF capacitors <u>in parallel</u> = 25 μF + 25 μF = 50 μF	[1] [1]
6(b)(ii)1.	Energy stored in the combination of 4 capacitors when pd reaches 540 V = ½ CV^2 = ½ x (Ans in (i)) x 540^2 =7.29 J Energy dissipated in one flash of light = 0.63 x 7.29 =4.6 J	[1,ecf] [1,ecf]
6(b)(ii)2.	Total energy remaining in all 4 capacitors after flash = 37% of 7.29 J = 2.69 J ½ CV_f^2 = 2.69 J, where C =Ans in (i) = 50 μF final pd across combination, V_f = 328 V Since 2 equal capacitors in series are of equal pd as their charge is the same, pd across each capacitor = ½ x V_f = ½ x 328 = 164 V	[1,ecf] [1,ecf]

7(a)(i)	Given: $\lambda = 4.5 \times 10^{-7}$ m; Intensity = 700 W m^{-2}; Area A = 1.0×10^{-4} m^2 Total energy of the beam of light, $E = \frac{Nhc}{\lambda}$ Intensity = P/A = E /At = $\frac{Nhc}{\lambda At}$ No. of photons per second, $N/t = \frac{\text{Intensity}(A)(\lambda)}{hc}$ $N/t = (700\ W\ m^{-2})(4.5 \times 10^{-7}\ m)(1.0 \times 10^{-4}\ m^2) / hc$ $N/t = 1.6$ or 1.58×10^{17}	[1] [1]
7(a)(ii)	Momentum of each photon, $p = h/\lambda$ $= h/ 4.5 \times 10^{-7}$ $= 1.5 \times 10^{-27}$ or 1.47×10^{-27} N s	[1] [1]

7(a)(iii)	Since the photons are all absorbed (ie not reflected), change in p for collision of ONE photon, $\Delta p = p_f - p_i$ $= p - 0$ = Ans in (ii) $=1.47 \times 10^{-27}$ N s change in p for collisions of ALL the photons in 1 s = (N/ t) (Δ p) = Ans in (i) x Ans in (ii) $= (1.58 \times 10^{17}) \times (1.47 \times 10^{-27})$ $= 2.3 \times 10^{-10}$ N	[1, ecf] [1, ecf]

7(b)(i)	$\Psi(x)$ = A sin 1kx at ground state when n = 1 Applying boundary condition at x = L, $\Psi(L) = 0$ since electron cannot be found at boundary with infinitely high potential A sin 1kL = 0 sin kL = 0 or A = 0 {trivial case} kL = nπ k = $\frac{\pi}{L}$ where n = 1	[1] [1]
7(b)(ii)	$\lvert\Psi\rvert^2$ is the probability density, or, probability per unit length (of finding the electron at x)	[1]
7(b)(iii)	Since the particle is confined within a well with infinite potential, the probability of finding the particle within the well = 1 ie 100% Substitute ψ: $\int_0^L A^2 \sin^2\left(\frac{1\pi x}{L}\right) dx = 1$ Since $\int_0^L A^2 \sin^2\left(\frac{\pi x}{L}\right) dx = Area\ \ under\ \ \lvert\psi(x)\rvert^2\text{-}x\ \ graph$ 1 = ($\frac{A^2}{2}$) x L { No further working needs to be shown } $\rightarrow$ A = $\sqrt{\frac{2}{L}}$	[1] [1]

8(a)(i)	SHM is an oscillatory motion (of an object) where the acceleration is (directly) proportional to displacement from the equilibrium position/a certain fixed pt & acceleration is always directed towards the equilibrium position/the fixed pt.	[1] [1]
8(a)(ii)	Deduce ½ period = 6.15 pm – 12 pm period T = 12.5 hours Deduce amplitude x_o = ½ (30 – 10) m x_0 = 10 m Deduce equation of motion is: x = $-x_o$ cos ωt where t = 0 corresponds to 12 pm $x = -10 \cos\frac{2}{12.5}t$ Marking: Allow ecf for values of x_0 & T  Since minimum depth required = 15 m, the earliest entrance is at depth of 5 m below equilibrium position where equilibrium position is at depth of 20 m, earliest entrance occurs at x = - 5 m Thus, $-5 = -10 \cos\frac{2}{12.5}t$ where t = time in hours after 12 noon $0.5 = \cos^{-1}\frac{2}{12.5}t$ t = 2.08 hrs = 2 hrs + 0.08(60) mins = 2 hrs + 4.8 mins = 2 hrs + 5 mins earliest time = 12 noon + 2 hrs and 5 mins = 2.05 pm	[1] [1] [1,ecf] [1] [1] [1, ecf]
8(a)(iii)	The tidal motion is a simple harmonic motion.	[1]
8(b)(i)1.	Period T = 20 ms {read-off the graph}	[1]

	f = 1/T = 50 Hz	[1]
8(b)(i)2.	Sketch of displ-position graph for fundamental mode of vibration}  Within each loop: <u>amplitude is different</u> for every point (except at nodes) Since max <u>acceleration</u>∝ <u>amplitude</u>, {do not accept: *displacement*} Max acceleration at all points is not the same. Marking: Award last mark only if first 2 marks are scored.	[1] [1] [1]
8(b)(ii)1.	resonance	[1]
8(b)(ii)2.	50 Hz (ecf, bi) corresponding to max amplitude { or equivalent }	[1,ecf]
8(b)(ii)3.	Damping is the effect due to a <u>gradual loss of energy</u> from an oscillating system <u>due to resistive forces</u> like friction or air resistance, which causes the <u>amplitude of the oscillations to decrease over time</u> 2nd graph: 1. approximately same (or slightly lower) resonant frequency & 2. sharpness of resonance reduced ie “obviously broader peak”. 3. smaller amplitude at every freq, (excluding pt 4); hence curves must not cross passes through same point on vertical-axis as original curve 4. passes through same point on vertical-axis as original curve {ie at f = 0}	[1] [1] [1] [1]

9(a)(i)	Draw a smooth curve/best fit line.	[1]
(ii)	Draw a horizontal line where the curve approaches the asymptote to obtain the background count rate.	[1]
(iii)	From the graph, background CR ~ 10.0 min^{-1}. CR on graph = 170 min^{-1} , (Corrected) CR due to source only = 170 – 10 = 160 min^{-1}. After $t_{1/2}$, (Corrected) CR due to source only = 160/2 = 80 min^{-1}, CR on graph = 80+10 = 90 min^{-1} , **For CR on graph to reduce from 170 to 90 min^{-1} ,** $t_{1/2}$ ~ 1.6 hours Need to find the half-life using another set of points from the graph, and then find the **average** half-life: CR on graph = 120 min^{-1} , (Corrected) CR due to source only = 120 – 10 = 110 min^{-1}. After $t_{1/2}$, (Corrected) CR due to source only = 110/2 = 55 min^{-1}, CR on graph = 55 + 10 = 65 min^{-1} , **For CR on graph to reduce from 120 to 65 min^{-1} ,** $t_{1/2}$ ~ 1.3 hours Hence averaged $t_{1/2}$ ~ 1.45 hours	[1] [1] [1]
(iv)	half-life: no change, because decay is spontaneous / is independent of environment / external factors like temperature.	[0] or [2]

(b)(i)	Total momentum before decay = total momentum after decay 0 = 4 (+V) + (A – 4](-v) -------- eqn (1) Marking: no mark if negative sign is omitted	[1]
(ii)	LHS = $\frac{\frac{1}{2}m_\alpha V^2}{\frac{1}{2}m_D v^2} = \frac{4V^2}{(A-4)v^2}$ -------- eqn (2), m_D= mass of daughter nucleus From eqn (1) in (b)(i), $\frac{V}{v} = \frac{A-4}{4}$ Sub eqn (1) into eqn (2), LHS = …. $= (\frac{1}{4}A - 1)$ (Shown) or KE = $\frac{p^2}{2m}$ {Allowed to recall & use} Thus $\frac{KE_{\alpha,i}}{KE_{D,i}} = \frac{p_\alpha^2}{2m_\alpha} \div \frac{p_D^2}{2m_D}$ where $p_\alpha = p_D$ {by PCLM} $= \frac{m_D}{m_\alpha} = \frac{(A-4)}{4} = \frac{A}{4} - 1$ (Shown)	[1] [1] [1]
(c)(i)	$^{212}_{83}Bi \rightarrow {}^{208}_{81}Tl + {}^{4}_{2}\alpha$ Energy released = $\Delta m c^2$= { 211.9459u – [207.9374*u* + 4.0015*u*] } c^2 = 0.007uc^2 {*u*: value taken from List of Data} = 1.0458 x 10^{-12} J = 6.54 MeV 1 MeV = 1.6 x 10^{-13} J	[1] [1] [1] [1]
(ii)	{Use (c)(i) & (b)(ii) to show energy of α-particle is 6.42 MeV} Let initial KE of α = x. Thus initial KE of daughter nucleus (i.e. Tl) = 6.54 – x MeV From (b)(ii), $\frac{x}{6.54 - x} = \frac{212}{4} - 1$ Solving, x = 6.42 MeV (shown)	[1] [1]

(d)(i)	Some of the decay energy is carried away as gamma radiation, so the alpha-particle has less kinetic energy.	[1]
(ii)	After decay, the total momentum of the α-particle, thallium nucleus & γ = 0 by conservation of linear momentum. For thallium nucleus & the α-particle to move off in opposite directions, all 3 particles will have to move off along the same line by the principle of conservation of momentum, which is not likely. Concept: Note that the γ -photon will therefore likely have a component of momentum in the direction perpendicular to the velocities indicated in the qn. PCLM must also hold for this perpendicular direction as well. The thallium nucleus and α-particle therefore are unlikely to move off in opposite directions.	[1] [1]